PREMIUM

Full-Length Mock Test

Mock 03

Engineering Physics

GATE PH

Total Questions	60
Maximum Marks	108
Duration	180 minutes (3 Hours)
Exam Pattern	MCQ + MSQ + NAT
Negative Marking	MCQ/MSQ: -0.33 per wrong answer
Designed For	EduNeuro Premium

Easy 20% | Moderate 33% | Hard 47%

Based on Previous Year Question Pattern Analysis | Generated by EduNeuro Premium

General Instructions

This question paper contains THREE sections: Section A, Section B, and Section C.

Section A: Multiple Choice Questions (MCQ) — carries one or two marks each. Only ONE option is correct.

Section B: Multiple Select Questions (MSQ) — carries two marks each. ONE OR MORE than one option(s) is/are correct.

Section C: Numerical Answer Type (NAT) — carries one or two marks. Enter the numerical value in the space provided.

Negative Marking: -0.33 marks will be deducted for each wrong MCQ and each wrong MSQ answer.

No negative marking for NAT questions.

There are 60 questions for a total of 100 marks.

Duration: 3 hours (180 minutes).

Use the space provided for rough work. Scribbling on the question paper is permitted.

Do not ask for clarification from the invigilator. Interpret questions as given.

Paper Structure

Section	Type	Questions	Marks/Q	Total Marks	Negative
A	MCQ	35	1 / 2	55	-0.33
B	MSQ	5	2	10	-0.33
C	NAT	20	1 / 2	35	None
Total		**60**		**100**	

Marking Scheme

- Correct MCQ: +1 mark (or +2 marks as specified per question)
- Wrong MCQ: -0.33 marks
- Correct MSQ: +2 marks
- Wrong MSQ (partial or no match): -0.33 marks
- Correct NAT: +1 mark (or +2 marks as specified per question)
- No answer / Unattempted: 0 marks
- NAT wrong answer: 0 marks (no negative marking)

Section A — Multiple Choice Questions (MCQ)

Q.1 [1 Mark] [EASY] [Quantum Mechanics — Operators]

Time-independent Schrödinger equation has:

A) First derivative
B) Second derivative
C) No derivative
D) Third derivative

Q.2 [1 Mark] [EASY] [Electrodynamics — EM Waves]

Wave-particle duality is fundamental to:

A) Classical mechanics
B) Quantum mechanics
C) Relativity
D) Thermodynamics

Q.3 [1 Mark] [EASY] [Mathematical Physics — Green Functions]

Time-independent Schrödinger equation has:

A) First derivative
B) Second derivative
C) No derivative
D) Third derivative

Q.4 [1 Mark] [EASY] [Optics — Polarization]

Wave-particle duality is fundamental to:

A) Classical mechanics
B) Quantum mechanics
C) Relativity
D) Thermodynamics

Q.5 [1 Mark] [EASY] [Quantum Mechanics — Wave Functions]

Time-independent Schrödinger equation has:

A) First derivative
B) Second derivative
C) No derivative
D) Third derivative

Q.6 [1 Mark] [EASY] [Electrodynamics — Maxwell Equations]

Wave-particle duality is fundamental to:

A) Classical mechanics
B) Quantum mechanics
C) Relativity
D) Thermodynamics

Q.7 [1 Mark] [EASY] [Mathematical Physics — Differential Equations]

Time-independent Schrödinger equation has:

A) First derivative

B) Second derivative

C) No derivative

D) Third derivative

Q.8 [1 Mark] [EASY] [Optics — Diffraction]

Wave-particle duality is fundamental to:

A) Classical mechanics
B) Quantum mechanics
C) Relativity
D) Thermodynamics

Q.9 [1 Mark] [EASY] [Quantum Mechanics — Wave Functions]

Time-independent Schrödinger equation has:

A) First derivative
B) Second derivative
C) No derivative
D) Third derivative

Q.10 [1 Mark] [EASY] [Electrodynamics — Potentials]

Wave-particle duality is fundamental to:

A) Classical mechanics
B) Quantum mechanics
C) Relativity
D) Thermodynamics

Q.11 [1 Mark] [EASY] [Mathematical Physics — Green Functions]

Time-independent Schrödinger equation has:

A) First derivative
B) Second derivative
C) No derivative
D) Third derivative

Q.12 [1 Mark] [EASY] [Optics — Geometric Optics]

Wave-particle duality is fundamental to:

A) Classical mechanics
B) Quantum mechanics
C) Relativity
D) Thermodynamics

Q.13 [2 Marks] [MODERATE] [Quantum Mechanics — Perturbation Theory]

Time-independent Schrödinger equation has:

A) First derivative
B) Second derivative
C) No derivative
D) Third derivative

Q.14 [2 Marks] [MODERATE] [Electrodynamics — Radiation]

Wave-particle duality is fundamental to:

A) Classical mechanics

B) Quantum mechanics

C) Relativity

D) Thermodynamics

Q.15 [2 Marks] [MODERATE] [Mathematical Physics — Green Functions]

Time-independent Schrödinger equation has:

A) First derivative
B) Second derivative
C) No derivative
D) Third derivative

Q.16 [2 Marks] [MODERATE] [Optics — Geometric Optics]

Wave-particle duality is fundamental to:

A) Classical mechanics
B) Quantum mechanics
C) Relativity
D) Thermodynamics

Q.17 [2 Marks] [MODERATE] [Quantum Mechanics — Wave Functions]

Time-independent Schrödinger equation has:

A) First derivative
B) Second derivative
C) No derivative
D) Third derivative

Q.18 [2 Marks] [MODERATE] [Electrodynamics — Potentials]

Wave-particle duality is fundamental to:

A) Classical mechanics
B) Quantum mechanics
C) Relativity
D) Thermodynamics

Q.19 [2 Marks] [MODERATE] [Mathematical Physics — Green Functions]

Time-independent Schrödinger equation has:

A) First derivative
B) Second derivative
C) No derivative
D) Third derivative

Q.20 [2 Marks] [MODERATE] [Optics — Geometric Optics]

Wave-particle duality is fundamental to:

A) Classical mechanics
B) Quantum mechanics
C) Relativity
D) Thermodynamics

Q.21 [2 Marks] [MODERATE] [Quantum Mechanics — Perturbation Theory]

Time-independent Schrödinger equation has:

A) First derivative

B) Second derivative

C) No derivative

D) Third derivative

Q.22 [2 Marks] [MODERATE] [Electrodynamics — EM Waves]

Wave-particle duality is fundamental to:

A) Classical mechanics
B) Quantum mechanics
C) Relativity
D) Thermodynamics

Q.23 [2 Marks] [MODERATE] [Mathematical Physics — Green Functions]

Time-independent Schrödinger equation has:

A) First derivative
B) Second derivative
C) No derivative
D) Third derivative

Q.24 [2 Marks] [MODERATE] [Optics — Polarization]

Wave-particle duality is fundamental to:

A) Classical mechanics
B) Quantum mechanics
C) Relativity
D) Thermodynamics

Q.25 [2 Marks] [MODERATE] [Quantum Mechanics — Perturbation Theory]

Time-independent Schrödinger equation has:

A) First derivative
B) Second derivative
C) No derivative
D) Third derivative

Q.26 [2 Marks] [MODERATE] [Electrodynamics — Radiation]

Wave-particle duality is fundamental to:

A) Classical mechanics
B) Quantum mechanics
C) Relativity
D) Thermodynamics

Q.27 [2 Marks] [MODERATE] [Mathematical Physics — Green Functions]

Time-independent Schrödinger equation has:

A) First derivative
B) Second derivative
C) No derivative
D) Third derivative

Q.28 [2 Marks] [MODERATE] [Optics — Diffraction]

Wave-particle duality is fundamental to:

A) Classical mechanics

B) Quantum mechanics

C) Relativity

D) Thermodynamics

Q.29 [2 Marks] [MODERATE] [Quantum Mechanics — Wave Functions]

Time-independent Schrödinger equation has:

A) First derivative
B) Second derivative
C) No derivative
D) Third derivative

Q.30 [2 Marks] [MODERATE] [Electrodynamics — Radiation]

Wave-particle duality is fundamental to:

A) Classical mechanics
B) Quantum mechanics
C) Relativity
D) Thermodynamics

Q.31 [2 Marks] [MODERATE] [Mathematical Physics — Differential Equations]

Time-independent Schrödinger equation has:

A) First derivative
B) Second derivative
C) No derivative
D) Third derivative

Q.32 [2 Marks] [MODERATE] [Optics — Polarization]

Wave-particle duality is fundamental to:

A) Classical mechanics
B) Quantum mechanics
C) Relativity
D) Thermodynamics

Q.33 [2 Marks] [HARD] [Quantum Mechanics — Schrödinger Equation]

Time-independent Schrödinger equation has:

A) First derivative
B) Second derivative
C) No derivative
D) Third derivative

Q.34 [2 Marks] [HARD] [Electrodynamics — Maxwell Equations]

Wave-particle duality is fundamental to:

A) Classical mechanics
B) Quantum mechanics
C) Relativity
D) Thermodynamics

Q.35 [2 Marks] [HARD] [Mathematical Physics — Differential Equations]

Time-independent Schrödinger equation has:

A) First derivative

B) Second derivative

C) No derivative

D) Third derivative

Section B — Multiple Select Questions (MSQ)

Q.36 [2 Marks] [HARD] [Optics — Polarization]

Which statements are correct?

A) Statement A
B) Statement B
C) Statement C
D) Statement D
E) Statement E

Q.37 [2 Marks] [HARD] [Quantum Mechanics — Wave Functions]

Which statements are correct?

A) Statement A
B) Statement B
C) Statement C
D) Statement D
E) Statement E

Q.38 [2 Marks] [HARD] [Electrodynamics — Potentials]

Which statements are correct?

A) Statement A
B) Statement B
C) Statement C
D) Statement D
E) Statement E

Q.39 [2 Marks] [HARD] [Mathematical Physics — Green Functions]

Which statements are correct?

A) Statement A
B) Statement B
C) Statement C
D) Statement D
E) Statement E

Q.40 [2 Marks] [HARD] [Optics — Geometric Optics]

Which statements are correct?

A) Statement A
B) Statement B
C) Statement C
D) Statement D
E) Statement E

Section C — Numerical Answer Type (NAT)

Q.51 [2 Marks] [HARD] [Mathematical Physics — Differential Equations]

Standard calculation using basic formula:

Answer: ____________________

Q.52 [2 Marks] [HARD] [Optics — Polarization]

Standard calculation using basic formula:

Answer: ____________________

Q.53 [2 Marks] [HARD] [Quantum Mechanics — Operators]

Standard calculation using basic formula:

Answer: ____________________

Q.54 [2 Marks] [HARD] [Electrodynamics — Radiation]

Standard calculation using basic formula:

Answer: ____________________

Q.55 [2 Marks] [HARD] [Mathematical Physics — Green Functions]

Standard calculation using basic formula:

Answer: ____________________

Q.56 [2 Marks] [HARD] [Optics — Diffraction]

Standard calculation using basic formula:

Answer: ____________________

Q.57 [2 Marks] [HARD] [Quantum Mechanics — Schrödinger Equation]

Standard calculation using basic formula:

Answer: ____________________

Q.58 [2 Marks] [HARD] [Electrodynamics — Maxwell Equations]

Standard calculation using basic formula:

Answer: ________________________

Q.59 [2 Marks] [HARD] [Mathematical Physics — Differential Equations]

Standard calculation using basic formula:

Answer: ____________________

Q.60 [2 Marks] [HARD] [Optics — Polarization]

Standard calculation using basic formula:

Answer: ____________________

Answer Key

Verify your answers after completing the full paper.

Q.1:	B	**Q.2:**	B	**Q.3:**	B
Q.4:	B	**Q.5:**	B	**Q.6:**	B
Q.7:	B	**Q.8:**	B	**Q.9:**	B
Q.10:	B	**Q.11:**	B	**Q.12:**	B
Q.13:	B	**Q.14:**	B	**Q.15:**	B
Q.16:	B	**Q.17:**	B	**Q.18:**	B
Q.19:	B	**Q.20:**	B	**Q.21:**	B
Q.22:	B	**Q.23:**	B	**Q.24:**	B
Q.25:	B	**Q.26:**	B	**Q.27:**	B
Q.28:	B	**Q.29:**	B	**Q.30:**	B
Q.31:	B	**Q.32:**	B	**Q.33:**	B
Q.34:	B	**Q.35:**	B	**Q.36:**	AC
Q.37:	AC	**Q.38:**	AC	**Q.39:**	AC
Q.40:	AC	**Q.41:**	AC	**Q.42:**	AC
Q.43:	AC	**Q.44:**	AC	**Q.45:**	AC
Q.46:	AC	**Q.47:**	AC	**Q.48:**	AC
Q.49:	AC	**Q.50:**	AC	**Q.51:**	10
Q.52:	10	**Q.53:**	10	**Q.54:**	10
Q.55:	10	**Q.56:**	10	**Q.57:**	10
Q.58:	10	**Q.59:**	10	**Q.60:**	10

Detailed Solutions

Question 1 *[Quantum Mechanics — Operators]*

Time-independent Schrödinger equation has:

Answer: (B)

Time-independent SE: - {"ó&Ò {#È + V<, Ò SÈ. Second-order PDE.

Question 2 *[Electrodynamics — EM Waves]*

Wave-particle duality is fundamental to:

Answer: (B)

Wave-particle duality is a core concept of quantum mechanics.

Question 3 *[Mathematical Physics — Green Functions]*

Time-independent Schrödinger equation has:

Answer: (B)

Time-independent SE: - {"ó&Ò {#È + V<, Ò SÈ. Second-order PDE.

Question 4 *[Optics — Polarization]*

Wave-particle duality is fundamental to:

Answer: (B)

Wave-particle duality is a core concept of quantum mechanics.

Question 5 *[Quantum Mechanics — Wave Functions]*

Time-independent Schrödinger equation has:

Answer: (B)

Time-independent SE: - {"ó&Ò {#È + V<, Ò SÈ. Second-order PDE.

Question 6 *[Electrodynamics — Maxwell Equations]*

Wave-particle duality is fundamental to:

Answer: (B)

Wave-particle duality is a core concept of quantum mechanics.

Question 7 *[Mathematical Physics — Differential Equations]*

Time-independent Schrödinger equation has:

Answer: (B)

Time-independent SE: - {"ó&Ò {#È + V<, Ò SÈ. Second-order PDE.

Question 8 *[Optics — Diffraction]*

Wave-particle duality is fundamental to:

Answer: (B)

Wave-particle duality is a core concept of quantum mechanics.

Question 9 *[Quantum Mechanics — Wave Functions]*

Time-independent Schrödinger equation has:

Answer:

(B)

Time-independent SE: - {"ó&Ò {#È + V<, Ò SÈ. Second-order PDE.

Question 10 *[Electrodynamics — Potentials]*

Wave-particle duality is fundamental to:

Answer: (B)

Wave-particle duality is a core concept of quantum mechanics.

Question 11 *[Mathematical Physics — Green Functions]*

Time-independent Schrödinger equation has:

Answer: (B)

Time-independent SE: - {"ó&Ò {#È + V<, Ò SÈ. Second-order PDE.

Question 12 *[Optics — Geometric Optics]*

Wave-particle duality is fundamental to:

Answer: (B)

Wave-particle duality is a core concept of quantum mechanics.

Question 13 *[Quantum Mechanics — Perturbation Theory]*

Time-independent Schrödinger equation has:

Answer: (B)

Time-independent SE: - {"ó&Ò {#È + V<, Ò SÈ. Second-order PDE.

Question 14 *[Electrodynamics — Radiation]*

Wave-particle duality is fundamental to:

Answer: (B)

Wave-particle duality is a core concept of quantum mechanics.

Question 15 *[Mathematical Physics — Green Functions]*

Time-independent Schrödinger equation has:

Answer: (B)

Time-independent SE: - {"ó&Ò {#È + V<, Ò SÈ. Second-order PDE.

Question 16 *[Optics — Geometric Optics]*

Wave-particle duality is fundamental to:

Answer: (B)

Wave-particle duality is a core concept of quantum mechanics.

Question 17 *[Quantum Mechanics — Wave Functions]*

Time-independent Schrödinger equation has:

Answer: (B)

Time-independent SE: - {"ó&Ò {#È + V<, Ò SÈ. Second-order PDE.

Question 18 *[Electrodynamics — Potentials]*

Wave-particle duality is fundamental to:

Answer:

(B)

Wave-particle duality is a core concept of quantum mechanics.

Question 19 *[Mathematical Physics — Green Functions]*

Time-independent Schrödinger equation has:

Answer: (B)

Time-independent SE: - {"ó&Ò {#È + V<, Ò SÈ. Second-order PDE.

Question 20 *[Optics — Geometric Optics]*

Wave-particle duality is fundamental to:

Answer: (B)

Wave-particle duality is a core concept of quantum mechanics.

Question 21 *[Quantum Mechanics — Perturbation Theory]*

Time-independent Schrödinger equation has:

Answer: (B)

Time-independent SE: - {"ó&Ò {#È + V<, Ò SÈ. Second-order PDE.

Question 22 *[Electrodynamics — EM Waves]*

Wave-particle duality is fundamental to:

Answer: (B)

Wave-particle duality is a core concept of quantum mechanics.

Question 23 *[Mathematical Physics — Green Functions]*

Time-independent Schrödinger equation has:

Answer: (B)

Time-independent SE: - {"ó&Ò {#È + V<, Ò SÈ. Second-order PDE.

Question 24 *[Optics — Polarization]*

Wave-particle duality is fundamental to:

Answer: (B)

Wave-particle duality is a core concept of quantum mechanics.

Question 25 *[Quantum Mechanics — Perturbation Theory]*

Time-independent Schrödinger equation has:

Answer: (B)

Time-independent SE: - {"ó&Ò {#È + V<, Ò SÈ. Second-order PDE.

Question 26 *[Electrodynamics — Radiation]*

Wave-particle duality is fundamental to:

Answer: (B)

Wave-particle duality is a core concept of quantum mechanics.

Question 27 *[Mathematical Physics — Green Functions]*

Time-independent Schrödinger equation has:

Answer:

(B)

Time-independent SE: - {"ó&Ò {#È + V<, Ò SÈ. Second-order PDE.

Question 28 *[Optics — Diffraction]*

Wave-particle duality is fundamental to:

Answer: (B)

Wave-particle duality is a core concept of quantum mechanics.

Question 29 *[Quantum Mechanics — Wave Functions]*

Time-independent Schrödinger equation has:

Answer: (B)

Time-independent SE: - {"ó&Ò {#È + V<, Ò SÈ. Second-order PDE.

Question 30 *[Electrodynamics — Radiation]*

Wave-particle duality is fundamental to:

Answer: (B)

Wave-particle duality is a core concept of quantum mechanics.

Question 31 *[Mathematical Physics — Differential Equations]*

Time-independent Schrödinger equation has:

Answer: (B)

Time-independent SE: - {"ó&Ò {#È + V<, Ò SÈ. Second-order PDE.

Question 32 *[Optics — Polarization]*

Wave-particle duality is fundamental to:

Answer: (B)

Wave-particle duality is a core concept of quantum mechanics.

Question 33 *[Quantum Mechanics — Schrödinger Equation]*

Time-independent Schrödinger equation has:

Answer: (B)

Time-independent SE: - {"ó&Ò {#È + V<, Ò SÈ. Second-order PDE.

Question 34 *[Electrodynamics — Maxwell Equations]*

Wave-particle duality is fundamental to:

Answer: (B)

Wave-particle duality is a core concept of quantum mechanics.

Question 35 *[Mathematical Physics — Differential Equations]*

Time-independent Schrödinger equation has:

Answer: (B)

Time-independent SE: - {"ó&Ò {#È + V<, Ò SÈ. Second-order PDE.

Question 36 *[Optics — Polarization]*

Which statements are correct?

Answer:

(A, C)

Based on theoretical analysis of the topic.

Question 37 *[Quantum Mechanics — Wave Functions]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 38 *[Electrodynamics — Potentials]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 39 *[Mathematical Physics — Green Functions]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 40 *[Optics — Geometric Optics]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 41 *[Quantum Mechanics — Perturbation Theory]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 42 *[Electrodynamics — Potentials]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 43 *[Mathematical Physics — Differential Equations]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 44 *[Optics — Geometric Optics]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 45 *[Quantum Mechanics — Perturbation Theory]*

Which statements are correct?

Answer:

(A, C)

Based on theoretical analysis of the topic.

Question 46 *[Electrodynamics — Maxwell Equations]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 47 *[Mathematical Physics — Differential Equations]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 48 *[Optics — Polarization]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 49 *[Quantum Mechanics — Operators]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 50 *[Electrodynamics — Radiation]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 51 *[Mathematical Physics — Differential Equations]*

Standard calculation using basic formula:

Answer: (10)

Apply the appropriate formula for the given data.

Question 52 *[Optics — Polarization]*

Standard calculation using basic formula:

Answer: (10)

Apply the appropriate formula for the given data.

Question 53 *[Quantum Mechanics — Operators]*

Standard calculation using basic formula:

Answer: (10)

Apply the appropriate formula for the given data.

Question 54 *[Electrodynamics — Radiation]*

Standard calculation using basic formula:

Answer:

(10)

Apply the appropriate formula for the given data.

Question 55

[Mathematical Physics — Green Functions]

Standard calculation using basic formula:

Answer: (10)

Apply the appropriate formula for the given data.

Question 56

[Optics — Diffraction]

Standard calculation using basic formula:

Answer: (10)

Apply the appropriate formula for the given data.

Question 57

[Quantum Mechanics — Schrödinger Equation]

Standard calculation using basic formula:

Answer: (10)

Apply the appropriate formula for the given data.

Question 58

[Electrodynamics — Maxwell Equations]

Standard calculation using basic formula:

Answer: (10)

Apply the appropriate formula for the given data.

Question 59

[Mathematical Physics — Differential Equations]

Standard calculation using basic formula:

Answer: (10)

Apply the appropriate formula for the given data.

Question 60

[Optics — Polarization]

Standard calculation using basic formula:

Answer: (10)

Apply the appropriate formula for the given data.